

AI Engineering

Building Applications with Foundation Models

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A joint embedding space that can represent data of different modalities is a *multimodal embedding space*. In a text–image joint embedding space, the embedding of an image of a man fishing should be closer to the embedding of the text “a fisherman” than the embedding of the text “fashion show”. This joint embedding space allows embeddings of different modalities to be compared and combined. For example, this enables text-based image search. Given a text, it helps you find images closest to this text.

AI as a Judge

The challenges of evaluating open-ended responses have led many teams to fall back on human evaluation. As AI has successfully been used to automate many challenging tasks, can AI automate evaluation as well? The approach of using AI to evaluate AI is called AI as a judge or LLM as a judge. An AI model that is used to evaluate other AI models is called an *AI judge*.¹⁵

While the idea of using AI to automate evaluation has been around for a long time,¹⁶ it only became practical when AI models became capable of doing so, which was around 2020 with the release of GPT-3. As of this writing, AI as a judge has become one of the most, if not the most, common methods for evaluating AI models in production. Most demos of AI evaluation startups I saw in 2023 and 2024 leveraged AI as a judge in one way or another. [LangChain’s State of AI](#) report in 2023 noted that 58% of

evaluations on their platform were done by AI judges. AI as a judge is also an active area of research.

Why AI as a Judge?

AI judges are fast, easy to use, and relatively cheap compared to human evaluators. They can also work without reference data, which means they can be used in production environments where there is no reference data.

You can ask AI models to judge an output based on any criteria: correctness, repetitiveness, toxicity, wholesomeness, hallucinations, and more. This is similar to how you can ask a person to give their opinion about anything. You might think, “But you can’t always trust people’s opinions.” That’s true, and you can’t always trust AI’s judgments, either. However, as each AI model is an aggregation of the masses, it’s possible for AI models to make judgments representative of the masses. With the right prompt for the right model, you can get reasonably good judgments on a wide range of topics.

Studies have shown that certain AI judges are strongly correlated to human evaluators. In 2023, [Zheng et al.](#) found that on their evaluation benchmark, MT-Bench, the agreement between GPT-4 and humans reached 85%, which is even higher than the agreement among humans (81%). AlpacaEval authors ([Dubois et al., 2023](#)) also found that their AI judges have a near

perfect (0.98) correlation with LMSYS's Chat Arena leaderboard, which is evaluated by humans.

Not only can AI evaluate a response, but it can also explain its decision, which can be especially useful when you want to audit your evaluation results. [Figure 3-7](#) shows an example of GPT-4 explaining its judgment.

Its flexibility makes AI as a judge useful for a wide range of applications, and for some applications, it's the only automatic evaluation option. Even when AI judgments aren't as good as human judgments, they might still be good enough to guide an application's development and provide sufficient confidence to get a project off the ground.

evaluate the quality of the following paragraph. give it a score from 1 to 5, and explain your decision.

Another drawback of this measurement is that higher lexical similarity scores don't always mean better responses. For example, on HumanEval, a code generation benchmark, OpenAI found that BLEU scores for incorrect and correct solutions were similar. This indicates that optimizing for BLEU scores isn't the same as optimizing for functional correctness (Chen et al., 2021).



I would rate the paragraph a **4.5 out of 5**.

Strengths:

1. **Clarity:** The paragraph clearly explains the drawback of using lexical similarity scores, providing a specific example.
2. **Evidence:** It effectively uses the HumanEval benchmark and OpenAI's findings to support the argument.
3. **Conciseness:** The paragraph is concise and directly to the point, making it easy to understand.

Areas for Improvement:

1. **Flow:** The transition between sentences is smooth, but the paragraph could benefit from a slightly stronger connection between the general statement and the specific example.

Figure 3-7. Not only can AI judges score, they also can explain their decisions.

How to Use AI as a Judge

There are many ways you can use AI to make judgments. For example, you can use AI to evaluate the quality of a response by itself, compare that response to reference data, or compare that response to another response. Here are naive example prompts for these three approaches:

1. Evaluate the quality of a response by itself, given the original question:

```
"Given the following question and answer, evaluate the quality of the answer for the question. Use the score from 1 to 5.
- 1 means very bad.
- 5 means very good.
Question: [QUESTION]
Answer: [ANSWER]
Score:"
```

2. Compare a generated response to a reference response to evaluate whether the generated response is the same as the reference response. This can be an alternative approach to human-designed similarity measurements:

```
"Given the following question, reference answer, and generated answer, evaluate whether this generated answer is the same as the reference answer.
Output True or False.
Question: [QUESTION]
Reference answer: [REFERENCE ANSWER]
Generated answer: [GENERATED ANSWER]"
```

3. Compare two generated responses and determine which one is better or predict which one users will likely prefer. This is helpful for generating preference data for post-training alignment (discussed in [Chapter 2](#)), test-

time compute (discussed in [Chapter 2](#)), and ranking models using comparative evaluation (discussed in the next section):

```
"Given the following question and two answers,  
better. Output A or B.  
Question: [QUESTION]  
A: [FIRST ANSWER]  
B: [SECOND ANSWER]  
The better answer is:"
```

A general-purpose AI judge can be asked to evaluate a response based on any criteria. If you're building a roleplaying chatbot, you might want to evaluate if a chatbot's response is consistent with the role users want it to play, such as "Does this response sound like something Gandalf would say?" If you're building an application to generate promotional product photos, you might want to ask "From 1 to 5, how would you rate the trustworthiness of the product in this image?" [Table 3-3](#) shows common built-in AI as a judge criteria offered by some AI tools.

Table 3-3. Examples of built-in AI as a judge criteria offered by some AI tools, as of September 2024. Note that as these tools evolve, these built-in criteria will change.

AI Tools	Built-in criteria
<u>Azure AI Studio</u>	Groundedness, relevance, coherence, fluency, similarity
<u>MLflow.metrics</u>	Faithfulness, relevance
<u>LangChain Criteria Evaluation</u>	Conciseness, relevance, correctness, coherence, harmfulness, maliciousness, helpfulness, controversiality, misogyny, insensitivity, criminality
<u>Ragas</u>	Faithfulness, answer relevance

It's essential to remember that AI as a judge criteria aren't standardized. Azure AI Studio's relevance scores might be very different from MLflow's relevance scores. These scores depend on the judge's underlying model and prompt.

How to prompt an AI judge is similar to how to prompt any AI application. In general, a judge's prompt should clearly explain the following:

1. The task the model is to perform, such as to evaluate the relevance between a generated answer and the question.

2. The criteria the model should follow to evaluate, such as “Your primary focus should be on determining whether the generated answer contains sufficient information to address the given question according to the ground truth answer”. The more detailed the instruction, the better.
3. The scoring system, which can be one of these:
 1. Classification, such as good/bad or relevant/irrelevant/neutral.
 2. Discrete numerical values, such as 1 to 5. Discrete numerical values can be considered a special case of classification, where each class has a numerical interpretation instead of a semantic interpretation.
 3. Continuous numerical values, such as between 0 and 1, e.g., when you want to evaluate the degree of similarity.

TIP

Language models are generally better with text than with numbers. It’s been reported that AI judges work better with classification than with numerical scoring systems.

For numerical scoring systems, discrete scoring seems to work better than continuous scoring. Empirically, the wider the range for discrete scoring, the worse the model seems to get. Typical discrete scoring systems are between 1 and 5.

Prompts with examples have been shown to perform better. If you use a scoring system between 1 and 5, include examples of what a response with a score of 1, 2, 3, 4, or 5 looks like, and if possible, why a response receives a certain score. Best practices for prompting are discussed in [Chapter 5](#).

Here's part of the prompt used for the criteria relevance by Azure AI Studio. It explains the task, the criteria, the scoring system, an example of an input with a low score, and a justification for why this input has a low score. Part of the prompt was removed for brevity.

Your task is to score the relevance between a generated answer and the question based on the ground truth answer in the range between 1 and 5, and please also provide the scoring reason.

Your primary focus should be on determining whether the generated answer contains sufficient information to address the given question according to the ground truth answer.

...

If the generated answer contradicts the ground truth answer, it will receive a low score of 1-2.

For example, for the question "Is the sky blue?" the ground truth answer is "Yes, the sky is blue." and the generated answer is "No, the sky is not blue."

In this example, the generated answer contradicts the ground truth answer by stating that the sky is not blue, when in fact it is blue.

This inconsistency would result in a low score of 1-2, and the reason for the low score would reflect the contradiction between the generated answer and the ground truth answer.

[Figure 3-8](#) shows an example of an AI judge that evaluates the quality of an answer when given the question.

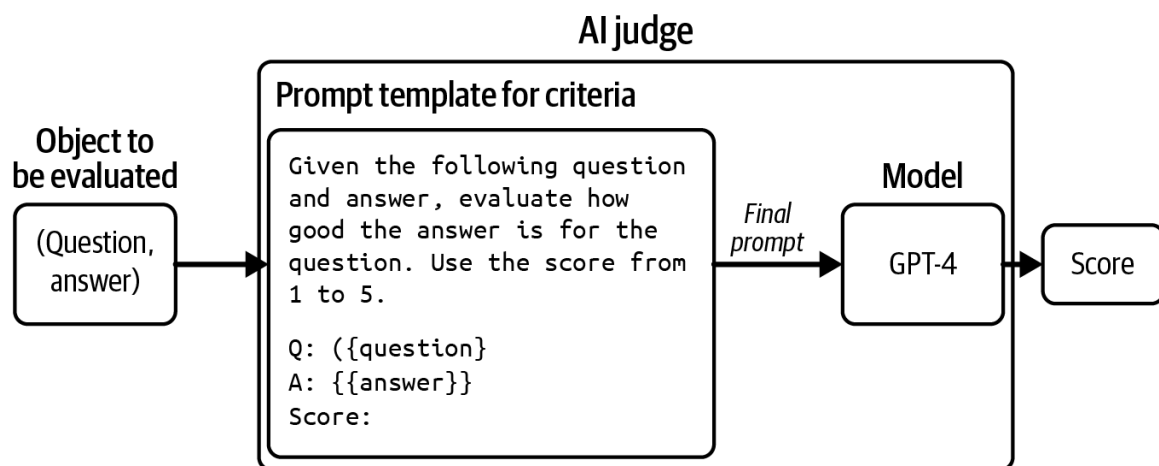


Figure 3-8. An example of an AI judge that evaluates the quality of an answer given a question.

An AI judge is not just a model—it's a system that includes both a model and a prompt. Altering the model, the prompt, or the model's sampling parameters results in a different judge.

Limitations of AI as a Judge

Despite the many advantages of AI as a judge, many teams are hesitant to adopt this approach. Using AI to evaluate AI seems tautological. The

probabilistic nature of AI makes it seem too unreliable to act as an evaluator. AI judges can potentially introduce nontrivial costs and latency to an application. Given these limitations, some teams see AI as a judge as a fallback option when they don't have any other way of evaluating their systems, especially in production.

Inconsistency

For an evaluation method to be trustworthy, its results should be consistent. Yet AI judges, like all AI applications, are probabilistic. The same judge, on the same input, can output different scores if prompted differently. Even the same judge, prompted with the same instruction, can output different scores if run twice. This inconsistency makes it hard to reproduce or trust evaluation results.

It's possible to get an AI judge to be more consistent. [Chapter 2](#) discusses how to do so with sampling variables. [Zheng et al. \(2023\)](#) showed that including evaluation examples in the prompt can increase the consistency of GPT-4 from 65% to 77.5%. However, they acknowledged that high consistency may not imply high accuracy—the judge might consistently make the same mistakes. On top of that, including more examples makes prompts longer, and longer prompts mean higher inference costs. In Zheng et al.'s experiment, including more examples in their prompts caused their GPT-4 spending to quadruple.

Criteria ambiguity

Unlike many human-designed metrics, AI as a judge metrics aren't standardized, making it easy to misinterpret and misuse them. As of this writing, the open source tools MLflow, Ragas, and LlamaIndex all have the built-in criterion *faithfulness* to measure how faithful a generated output is to the given context, but their instructions and scoring systems are all different. As shown in [Table 3-4](#), MLflow uses a scoring system from 1 to 5, Ragas uses 0 and 1, whereas LlamaIndex's prompt asks the judge to output YES and NO.